

YOUR OPTIMAL RANGES

Testosterone by age: normal isn't the same as optimal

The standard lab range spans every man from 19 to 80, so a result can sit comfortably “in range” yet be low for your age — the zone where many men feel flat despite a normal report. Below is the functional target for each decade.

AGE	STANDARD LAB RANGE	FUNCTIONAL OPTIMAL	WHAT IT MEANS
19–29	264–1,200	700–950	Peak years. A low reading here warrants proper investigation.
30–39	264–1,100	600–850	The gradual ~1%/year decline begins. Lifestyle shows most.
40–49	264–1,000	500–750	Symptoms often appear. Free testosterone matters more.
50–59	264–950	400–650	SHBG rises — always test free T alongside total.
60+	264–900	350–550	A “normal” total can still be suboptimal for quality of life.

Values in ng/dL. Multiply by 0.03467 for nmol/L. Lower bound = harmonised “low” threshold (Endocrine Society); optimal ranges from The Testosterone Blueprint, Ch. 2.

BEYOND TOTAL TESTOSTERONE

Free T

Optimal **15–30** ng/dL

The ~2% that's active. Calculated from total T, SHBG & albumin — the real story when SHBG is high.

SHBG

Range **16–57** nmol/L

Rises with age (50+: 20–82). High SHBG lowers free T; low SHBG can flag metabolic issues.

Oestradiol

Optimal **20–30** pg/mL

Men need it too — for bone, libido & mood. Ask for a sensitive/LC-MS assay for accuracy.

BEFORE YOU BOOK

Your blood-test checklist

A “normal” total testosterone alone tells you less the older you get. Take this list to your GP or private lab so you test the right markers, the right way, the first time.

WHAT TO TEST

- ☐ **Total testosterone** — morning sample
- ☐ **SHBG** — binding globulin
- ☐ **Albumin** — needed to calculate free T
- ☐ **Free testosterone** — calculated (Vermeulen)
- ☐ **Oestradiol** — sensitive / LC-MS assay
- ☐ **LH & FSH** — pituitary signals
- ☐ **Prolactin** — if libido is low

WORTH ADDING

- ☐ **TSH** — thyroid, mimics low-T
- ☐ **HbA1c or fasting glucose** — insulin health
- ☐ **Lipid panel** — cardiovascular baseline
- ☐ **Vitamin D** — often low, easily fixed
- ☐ **Ferritin & full blood count** — energy & recovery
- ☐ **Your lab's own ranges** — note them; they vary

HOW TO TEST RIGHT

- ☐ **Sample between 7–10am.** Testosterone peaks in the morning; an afternoon reading can run 20–25% lower.
- ☐ **Fasted & well-rested.** Avoid testing after heavy alcohol or a poor night's sleep.
- ☐ **Not during acute illness.** Infection and stress temporarily suppress levels.
- ☐ **Confirm a low result.** Never act on a single reading — repeat before any conclusion.
- ☐ **Insist on a sensitive oestradiol assay.** Standard assays misread male-range levels.
- ☐ **Interpret together.** Read total, free T and SHBG as one picture, alongside symptoms.

PUT IT TO WORK

Plot your results against your age — free & interactive.

Tool → testosteroneblueprintguide.com/testosterone-levels-by-age · Full method → [The Testosterone Blueprint](#)